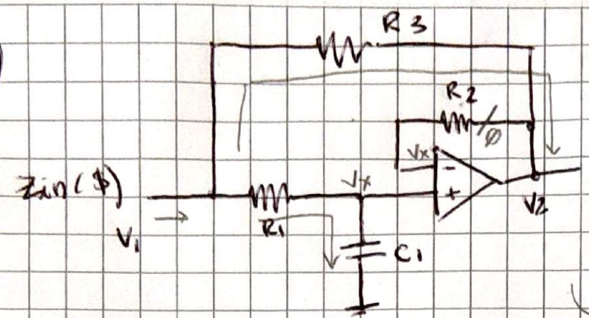
2b)



$$\rightarrow \frac{V_{in} - V_x}{R_1} = \frac{V_x - 0}{\frac{1}{\$C_1}} = V_x \$C_1$$

$$\rightarrow \frac{V_0 - V_x}{R_2} = 0 \Big\} \; \underline{\underline{V_0 = V_x}}$$

$$\frac{V_{in} - V_0}{R_1} = V_0 \$C_1$$

$$V_{in} - V_0 = V_0 \$C_1 R_1$$

$$V_{in} = V_0(\$C_1R_1 + 1)$$

$$\underline{\underline{V_0 = \frac{V_{in}}{\$C_1R_1 + 1}}}$$

$$\rightarrow I_{in} = \frac{V_i - V_x}{R_1} + \frac{V_i - V_0}{R_3}$$

$$I_{in} = \frac{V_i - V_0}{R_1} + \frac{V_i - V_0}{R_3}$$

$$I_{in} = V_i\left(\frac{1}{R_1} + \frac{1}{R_3}\right) - V_0\left(\frac{1}{R_1} + \frac{1}{R_3}\right)$$

$$I_{in} = V_i \frac{R_3 + R_1}{R_1R_3} - \frac{V_i}{\$C_1R_1 + 1} \cdot \frac{R_3 + R_1}{R_3R_1}$$

$$I_{in} = \frac{V_i(R_3 + R_1)}{R_1R_3}\left(1 - \frac{1}{\$C_1R_1 + 1}\right) = V_i \frac{(R_3 + R_1)}{R_1R_3} \cdot \frac{\$C_1R_1 \cancel{+1} \cancel{-1}}{\$C_1R_1 + 1}$$

$$I_{in} = V_{in} \cdot \frac{(R_3 + R_1) \cdot \$C_1\cancel{R_1}}{(\$C_1R_1 + 1)\cancel{R_1}R_3} = V_{in}.$$

$$\frac{V_{in}}{I_{in}} = Z_{in} = \frac{(\$C_1R_1 + 1)R_3}{(R_3 + R_1)\$C_1} = \frac{\$C_1R_1R_3 + R_3}{\$(C_1R_1 + C_1R_3)} = \frac{C_1R_1R_3}{C_1R_1 + C_1R_3} + \frac{R_3}{\$(C_1R_1 + C_1R_3)}$$